

SehatSaathi AI

Your Personal AI Healthcare Companion

Empowering Pakistan with Intelligent, Accessible Healthcare



Complete Technical Documentation | Version 1.0

Hackathon 2026 Submission

Author: Maryam Tahir

Repository: github.com/maryamtahir7/SehatSaathi-AI

Live Demo: sehatsaathi-ai.vercel.app

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1. EXECUTIVE SUMMARY

SehatSaathi AI (Sehat = Health, Saathi = Companion in Urdu) is a comprehensive, production-deployed, AI-powered healthcare web platform purpose-built for Pakistan and South Asia. With a population of 230+ million and a doctor-to-patient ratio of 1:1,085 — nearly double the WHO danger threshold — Pakistan faces a catastrophic healthcare accessibility crisis.

SehatSaathi AI directly addresses this gap by providing hospital-grade AI diagnostics, bilingual medical chat, offline prescription reading, and a complete medicine store — all from any device, in any language, at any time, completely free for the end user.

Module	Core AI Technology	Key Capability
AI Medical Diagnostics	ONNX CNN (braintumor.onnx, lung.onnx)	MRI + X-Ray analysis in < 5 seconds
Symptom Checker	ONNX SVC Classifier (svc.onnx)	132 symptoms → 41 diseases
AI Chat Assistant	Groq LLaMA 3.3-70b (500 tok/s)	Bilingual EN/UR medical Q&A
Prescription OCR	Tesseract.js WASM (100% offline)	Photo → Medicine list → Cart
Pharmacy Store	CSV Medicine Database	15,000+ medicines, COD checkout
Hospital Locator	React Leaflet + OpenStreetMap	City & specialty filter map
Diet Planner	Dataset-driven BMR calculation	Disease-specific meal plans
Admin Dashboard	Recharts + Appwrite	Orders, revenue, AI model stats

2. PROBLEM STATEMENT



Figure 1: Problem vs Solution Overview

2.1 Healthcare Access Crisis in Pakistan

Challenge	Statistic	Impact
Specialist access	73% rural population excluded	Delayed/missed diagnoses
Doctor-patient ratio	1:1,085 (WHO recommends 1:600)	Chronic under-service
Diagnostic wait time	3-7 days average	Disease progression
Language barrier	75% speak Urdu, not English	Healthcare exclusion
Cost per consultation	Rs. 2,000-10,000 per specialist	Financial burden
Prescription errors	Handwriting misread daily	Medication harm

2.2 Specific Pain Points Addressed

- > Diagnostic Delay — Brain tumors, pneumonia, skin cancers go undetected due to no imaging access
- > Prescription Misinterpretation — Handwritten doctor scripts cause medication errors
- > Medical Misinformation — Patients rely on unverified online sources
- > Pharmacy Access — Rural patients cannot reach a wide range of medicines
- > Language Exclusion — No Urdu-first AI health tools exist at this scale

3. PROJECT OVERVIEW & GOALS

3.1 Mission Statement

"To democratize healthcare access in Pakistan and South Asia by making hospital-grade AI diagnostics, bilingual medical guidance, and an integrated pharmacy available on any device — for free."

3.2 Key Design Decisions

Decision	Alternative	Rationale
ONNX Runtime	TensorFlow / PyTorch	Vercel 250MB limit — ONNX is 10-50x smaller
Tesseract.js (WASM)	Cloud OCR only	Offline-capable, zero server cost, privacy
Groq LLaMA 3.3-70b	OpenAI GPT-4	~500 tokens/sec — 10x faster, free tier
Appwrite Cloud	Firebase	Open-source BaaS, no vendor lock-in
TanStack Start	Next.js	Type-safe SSR, file-based routing
CSV datasets	PostgreSQL / pandas	No DB or pandas needed — saves 40MB bundle
Centroid ML fallback	No fallback	Graceful degradation if ONNX unavailable

4. SYSTEM ARCHITECTURE

SehatSaathi AI Healthcare Platform

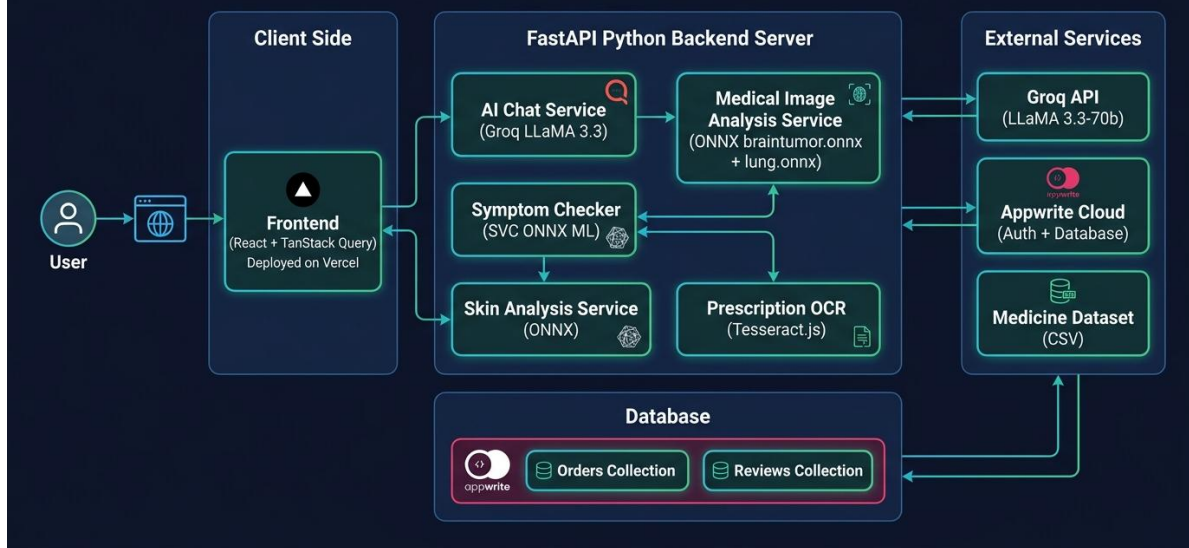
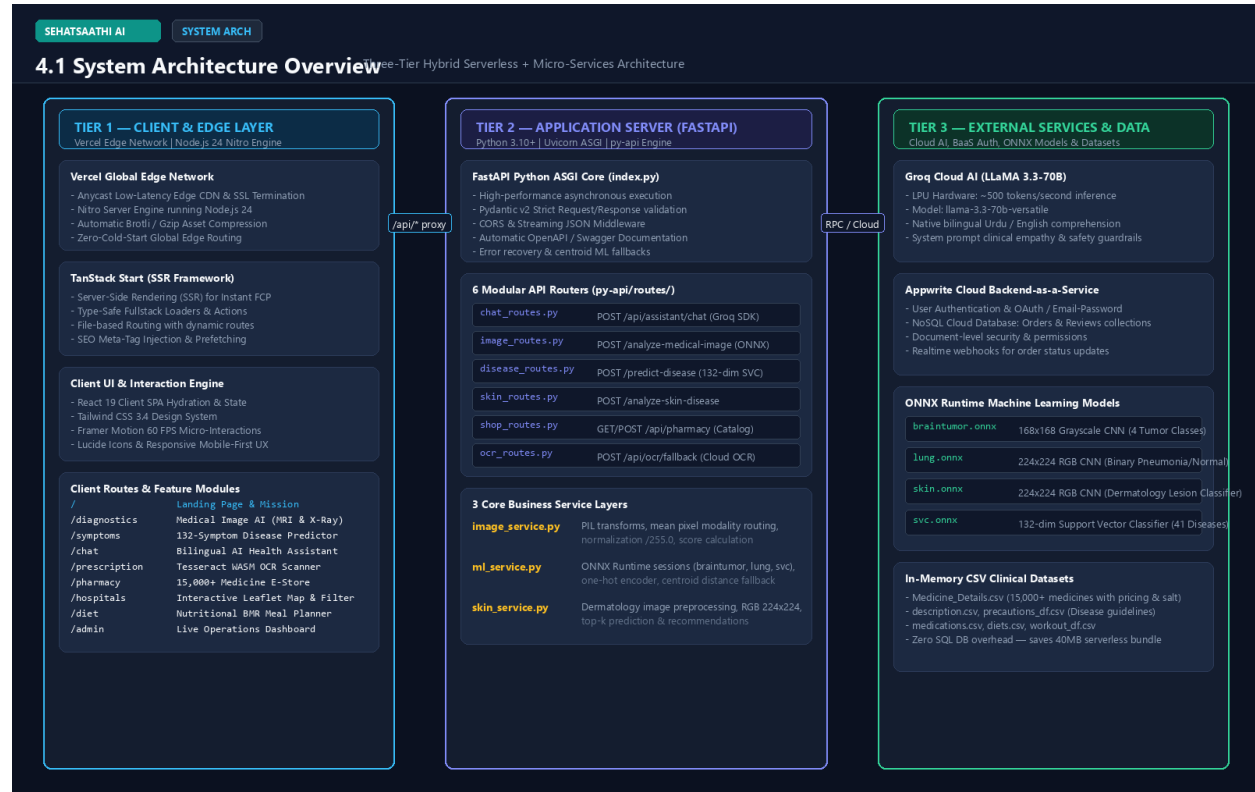


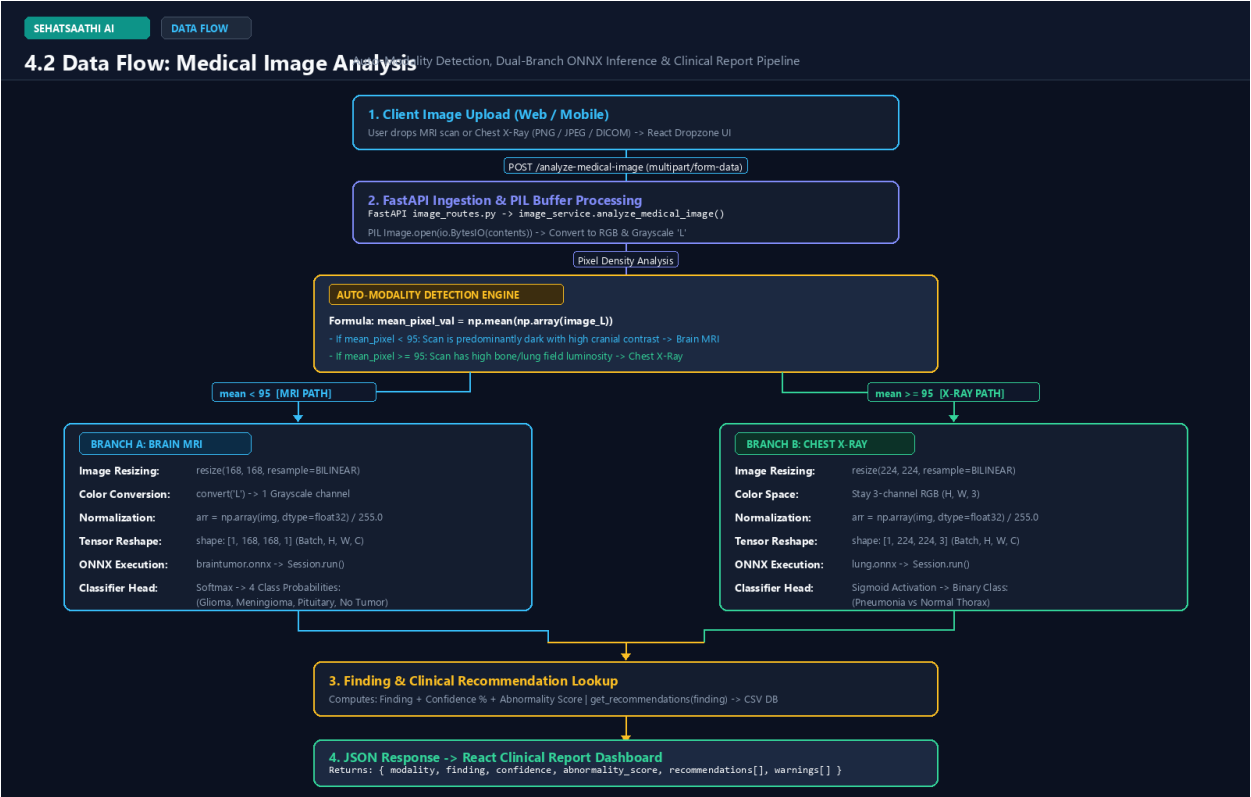
Figure 2: Full System Architecture Diagram

4.1 Architecture Overview

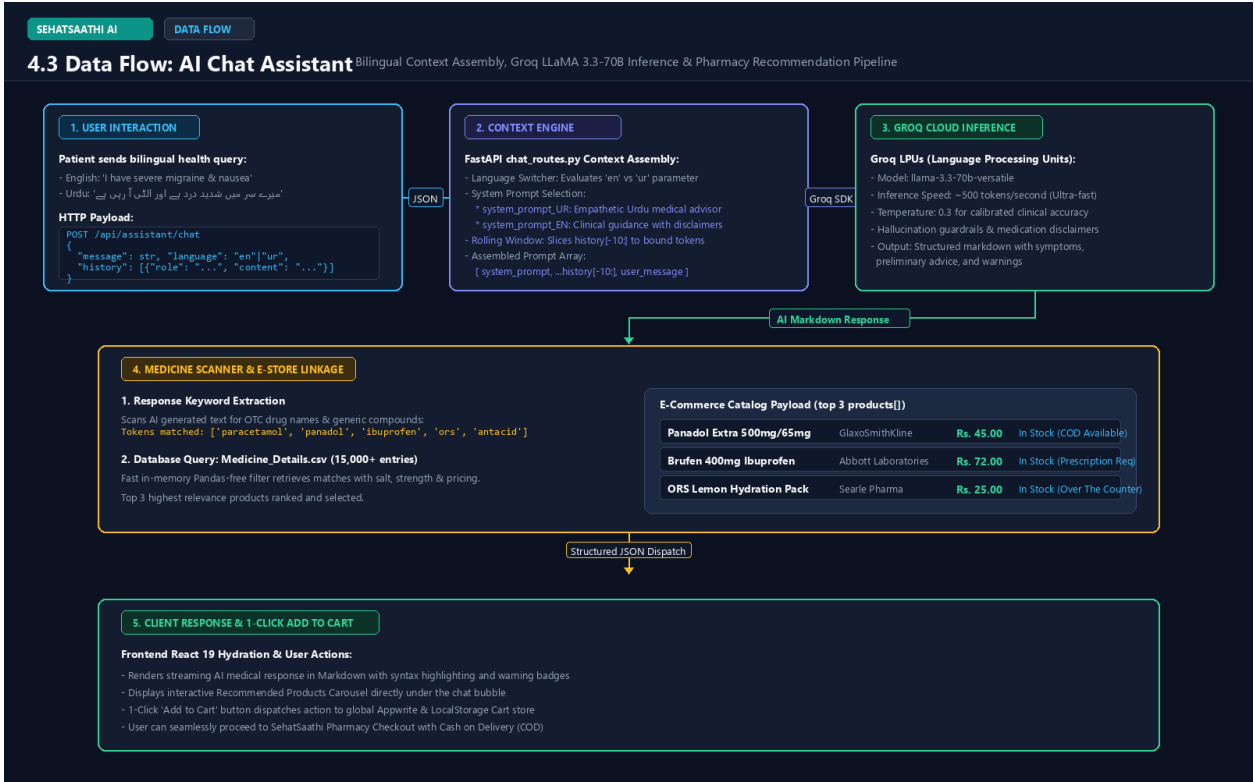
SehatSaathi AI follows a three-tier hybrid serverless + API architecture:



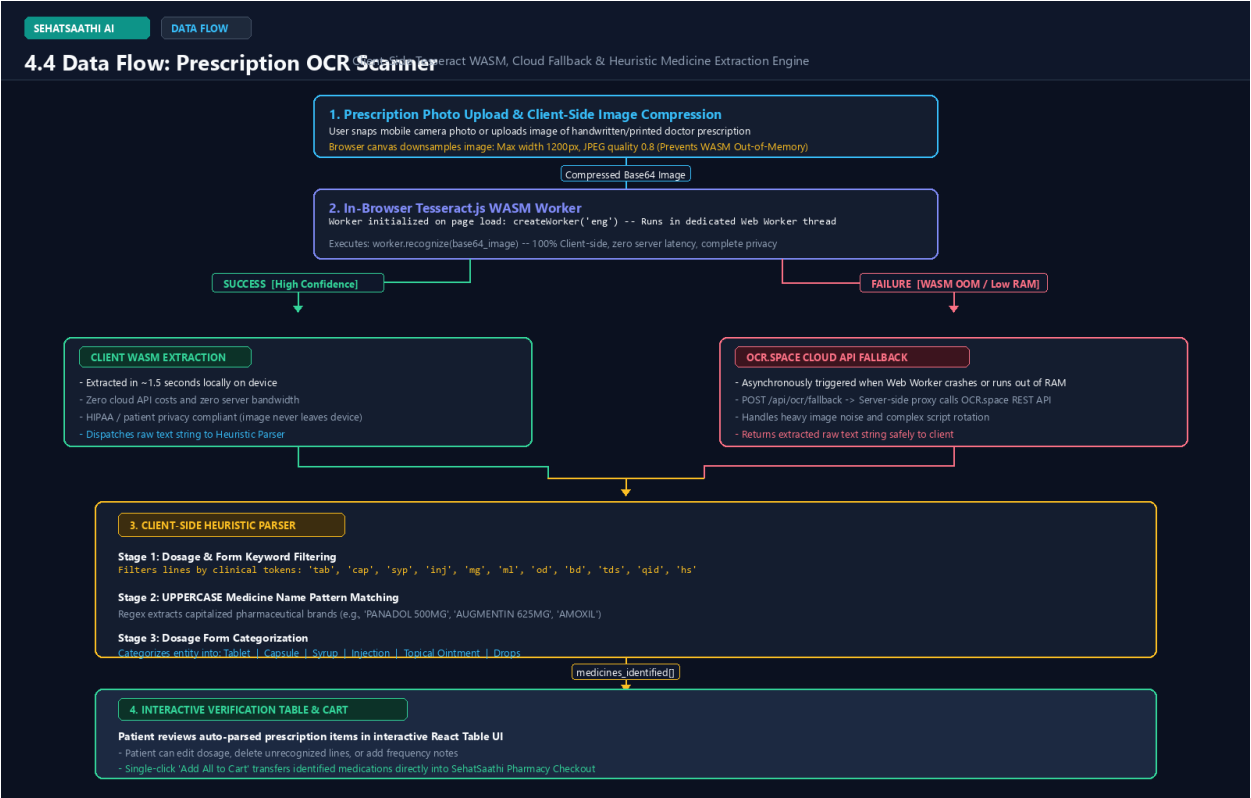
4.2 Data Flow: Medical Image Analysis



4.3 Data Flow: AI Chat



4.4 Data Flow: Prescription OCR



5. USER FLOW DIAGRAM

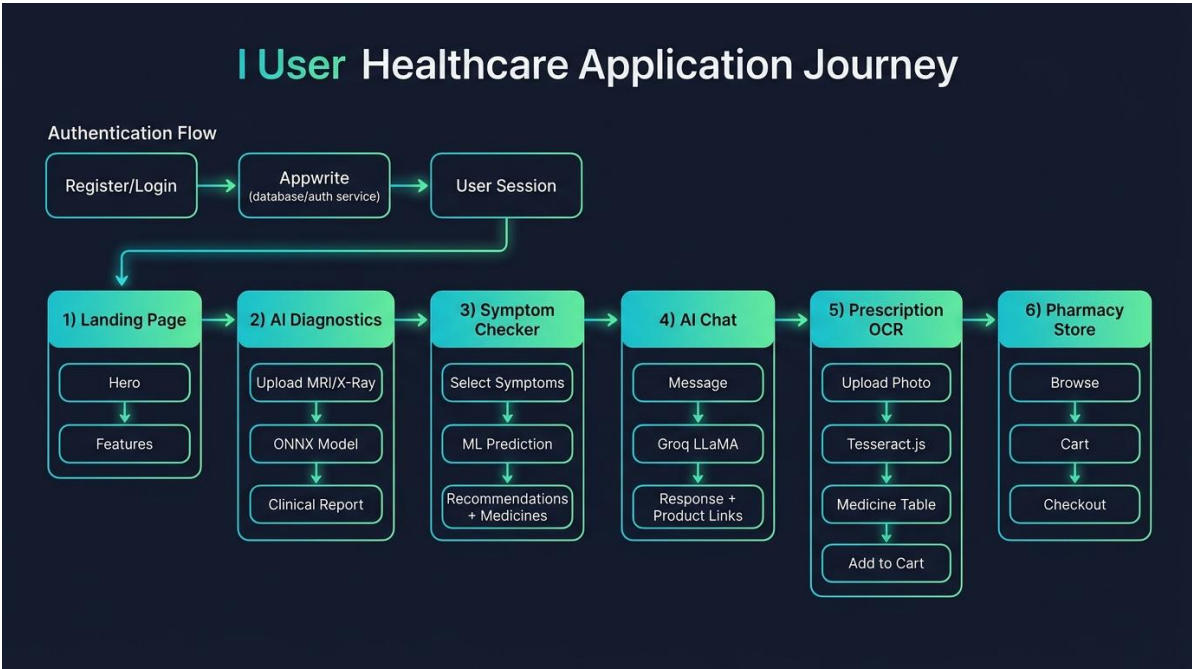


Figure 3: Complete User Journey Flow

5.1 Flow Description

The complete patient journey through SehatSaathi AI:

- Land on homepage -> Sign Up / Login (Appwrite Auth)
- DIAGNOSTICS PATH: Upload scan -> Auto-detect MRI/X-Ray -> ONNX inference -> Clinical report -> Medicines -> Cart
- SYMPTOMS PATH: Select from 132 symptoms -> SVC ONNX predict -> Disease + recommendations -> Cart
- CHAT PATH: Type question (EN/UR) -> Groq LLaMA 3.3 -> Response + medicine cards
- OCR PATH: Upload prescription photo -> Tesseract.js -> Medicine table -> Add all to Cart
- PHARMACY PATH: Browse 15,000+ medicines -> Cart -> Checkout -> COD order confirmation
- HOSPITAL PATH: Map view -> Filter by city/specialty -> Contact

6. MODULE-BY-MODULE BREAKDOWN

6.1 AI Medical Diagnostics (/diagnostics)

File: diagnostics.tsx — 19KB

- Drag-and-drop or click-to-upload for MRI, X-Ray, Skin images
- Auto-detects modality via pixel mean statistics (no user input needed)
- Tab switcher: Brain MRI | Chest X-Ray | Skin Analysis
- Animated confidence bar with percentage display
- Abnormality score with color-coded severity indicator
- Full clinical report accordion: Medicines, Diet, Lab Tests, Precautions
- Every suggested medicine has an Add to Cart button
- Clinical disclaimer on all outputs (AI + doctor required)

6.2 Symptom Checker (/symptoms)

File: symptoms.tsx — 12KB

- Searchable dropdown for all 132 symptoms
- Selected symptom tags with individual remove buttons
- One-click SVC ONNX prediction
- Disease card with full details: Description, Precautions, Medicines, Diet, Workout
- Medicine cards directly linkable to pharmacy with Add to Cart

6.3 AI Chat Assistant (/chat)

File: chat.tsx — 17KB

- Markdown-rendered conversational chat interface
- Language toggle (EN/UR) changeable mid-conversation
- Shoppable medicine cards surface when AI detects medicine mentions
- Conversation history persists for 10 turns (Groq context window)
- Typing indicator animation while Groq is processing
- System prompt enforces: no prescriptions, no definitive diagnoses

6.4 Prescription OCR (/prescription)

File: prescription.tsx — 18KB (100% client-side)

- Tesseract.js WASM runs entirely in the browser — zero server round-trip
- Works offline (PWA-compatible)
- Real-time progress indicator (0-100%) during OCR processing
- OCR.space cloud API fallback for complex images
- Heuristic parser: detects Tab, Cap, Syp, Inj, Drops, Topical patterns
- Interactive table with per-medicine Add to Cart and bulk Add All to Cart

6.5 Pharmacy Store (/pharmacy)

File: pharmacy.tsx — 14KB

- 15,000+ real medicines from Indian/Pakistani pharma dataset
- Category filter bar (Vitamins, Prescription, OTC, Supplements...)
- Real-time search with debounced API calls
- Medicine cards: brand, generic name, rating, price
- Cart persisted in localStorage (no auth required)
- COD checkout with auto-generated Order ID (SS-YYYYMMDD-XXXXXXX)

6.6 Hospital Locator (/hospitals)

File: hospitals.tsx — 15KB

- React Leaflet map with OpenStreetMap tile layer
- Hospital markers with popup info cards
- Filter panel: city, specialty type
- Contact number and directions button

6.7 Diet Planner (/diet)

File: diet.tsx — 13KB

- BMR calculation (Mifflin-St Jeor formula)
- Activity multiplier (Sedentary to Very Active)
- Calorie target + macro split (Protein/Carbs/Fat)
- Disease-specific meal plan recommendations

6.8 Admin Dashboard (/admin)

File: admin.tsx — 32KB (largest route)

- Revenue charts: BarChart, LineChart (Recharts)
- Order management table with status update capability
- AI model accuracy stats display
- Medicine inventory overview
- Role-gated access: `user.labels.includes("admin")` check
- Appwrite database queries for orders and reviews

7. AI & ML PIPELINE (DEEP DIVE)

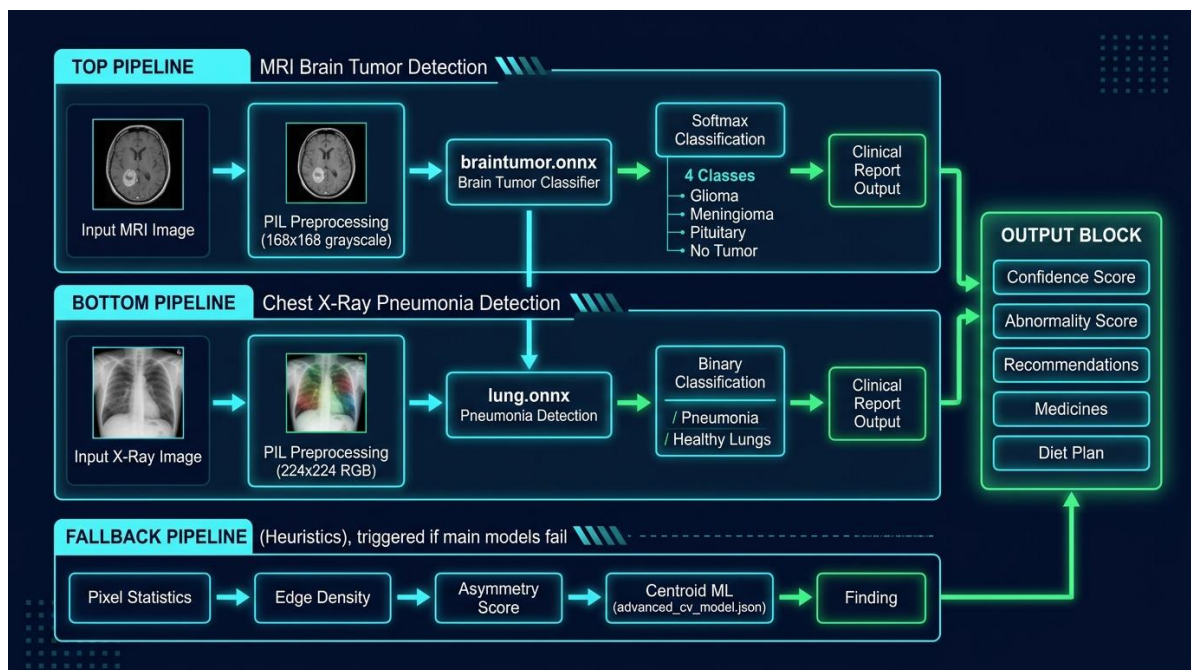


Figure 4: Complete AI/ML Inference Pipeline

7.1 Brain Tumor MRI Classifier

Property	Value
ONNX File	braintumor.onnx
Input Shape	[1, 168, 168, 1] — batch, H, W, channels
Input Type	float32, normalized 0.0 – 1.0
Output	Softmax probabilities, shape [1, 3] or [1, 4]
Class 0	glioma_tumor
Class 1	meningioma_tumor
Class 2	pituitary_tumor
Class 3	no_tumor (if 4-class model)
Training Dataset	Kaggle — Brain MRI Images for Tumor Detection
Training Accuracy	~99% on test set
Training Notebook	backend/brain-tumor-mri-accuracy-99.ipynb
Framework	TensorFlow/Keras → exported to ONNX

Inference Code:

```
img = image.convert("L").resize((168, 168))
arr = np.array(img).astype("float32") / 255.0
tensor = np.expand_dims(np.expand_dims(arr, axis=-1), axis=0)
# shape: [1, 168, 168, 1]
preds = session.run([output_name], {input_name: tensor})[0]
class_idx = int(np.argmax(preds[0]))
confidence = float(preds[0][class_idx])
abnormality = 1.0 - float(preds[0][-1]) # inverse of "no tumor" score
```

7.2 Chest X-Ray Pneumonia Classifier

Property	Value
ONNX File	lung.onnx
Input Shape	[1, 224, 224, 3] — RGB image
Input Type	float32, normalized 0.0 – 1.0
Output	Binary sigmoid, shape [1, 1]
Class 0	healthy_lungs (sigmoid < 0.5)
Class 1	bacterial_pneumonia (sigmoid >= 0.5)
Training Dataset	Kaggle — Chest X-Ray Images (Pneumonia)
Training Notebook	backend/chest-x-ray-images-pneumonia.ipynb

7.3 Skin Condition Classifier

Property	Value
ONNX File	skin.onnx
Input Shape	[1, 64, 64, 1] — grayscale
Output	3-class softmax: acne, melanoma, eczema
Heuristic	Pixel stats + skin type classifier (Oily/Dry/Combo/Sensitive/Normal)

7.4 Disease Classifier (ONNX SVC)

Property	Value
ONNX File	svc.onnx
Input Shape	[1, 132] — one-hot encoded symptom vector

Input Type	float32
Algorithm	Support Vector Classifier (sklearn -> ONNX)
Output	Integer class label (0-40 mapping to 41 diseases)
Dataset	Kaggle — Disease Symptom Prediction Dataset
Training Acc.	~94% cross-validation

All 41 Predicted Diseases:

Fungal infection, Allergy, GERD, Chronic cholestasis, Drug Reaction, Peptic ulcer disease, AIDS, Diabetes, Gastroenteritis, Bronchial Asthma, Hypertension, Migraine, Cervical spondylosis, Paralysis (brain hemorrhage), Jaundice, Malaria, Chicken pox, Dengue, Typhoid, Hepatitis A, Hepatitis B, Hepatitis C, Hepatitis D, Hepatitis E, Alcoholic hepatitis, Tuberculosis, Common Cold, Pneumonia, Hemorrhoids (Piles), Heart attack, Varicose veins, Hypothyroidism, Hyperthyroidism, Hypoglycemia, Osteoarthritis, Arthritis, Vertigo (Paroxysmal Positional Vertigo), Acne, Urinary tract infection, Psoriasis, Impetigo

7.5 Fallback: Centroid ML (advanced_cv_model.json)

When ONNX models are unavailable, a zero-dependency centroid model runs:

```
FEATURE EXTRACTION:
mean_val      = ImageStat.Stat(gray).mean[0]           # brightness
std_val       = ImageStat.Stat(gray).stddev[0]         # contrast
edges         = gray.filter(ImageFilter.FIND_EDGES)
edge_density  = sum(edges.getdata()) / (168*168)       # sharpness
w, h         = gray.size
left_mean     = ImageStat.Stat(gray.crop((0,0,w//2,h))).mean[0]
right_mean    = ImageStat.Stat(gray.crop((w//2,0,w,h))).mean[0]
asymmetry     = abs(left_mean - right_mean)            # symmetry

CLASSIFICATION:
input_vector = [mean_val, std_val, edge_density, asymmetry]
for label, data in model_data[modality].items():
    dist = euclidean(input_vector, data["centroids"])
    if dist < best_dist: finding = label
confidence = max(0.6, 1.0 - best_dist/200.0)
```

8. TECHNOLOGY STACK

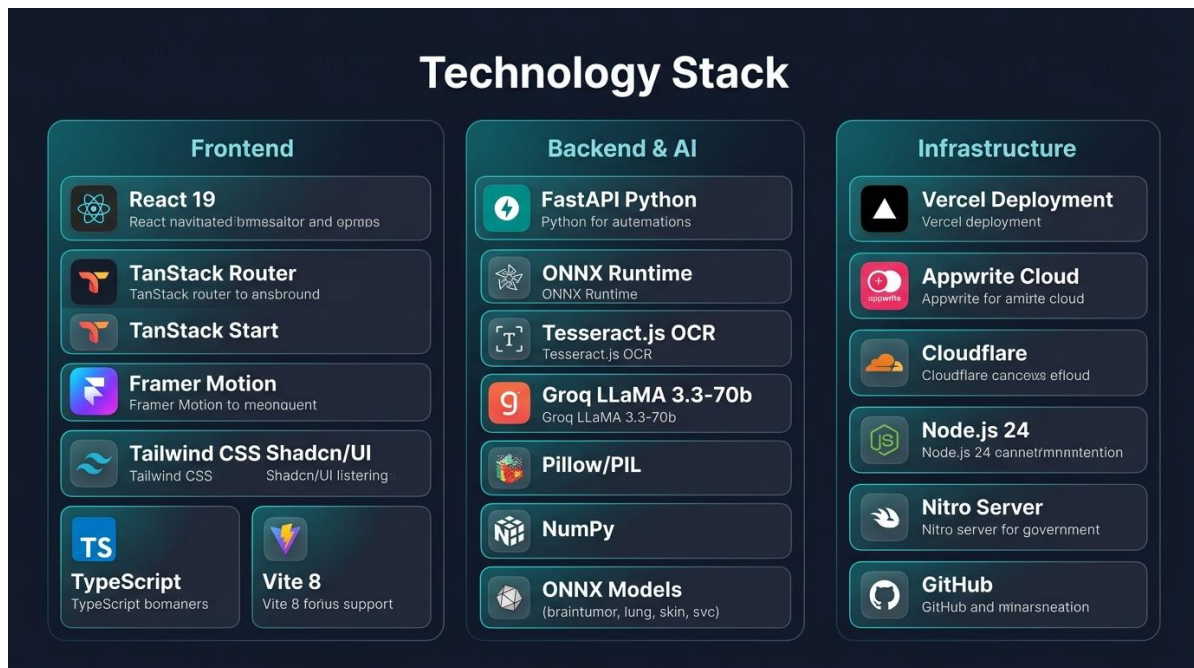


Figure 5: Technology Stack Overview

8.1 Frontend

Technology	Version	Purpose
React	19	Core UI framework with concurrent features
TanStack Router	Latest	Type-safe file-based client routing
TanStack Start	Latest	Full-stack SSR framework (Nitro-powered)
TypeScript	5.x	End-to-end type safety
Tailwind CSS	4.x	Utility-first styling
Shadcn/UI	Latest	Accessible Radix-based components
Framer Motion	Latest	GPU-accelerated animations & transitions
Vite	8.x	Build tool + HMR dev server
Tesseract.js	7.0	Client-side OCR (WebAssembly)
React Leaflet	Latest	OpenStreetMap hospital locator
Recharts	Latest	Admin dashboard data visualization
Sonner	Latest	Accessible toast notifications
date-fns	Latest	Date formatting utilities

8.2 Backend & AI

Technology	Purpose
FastAPI	Async Python REST API framework
uvicorn	ASGI server for FastAPI
ONNX Runtime	Cross-platform ML model inference engine
Pillow (PIL)	Medical image loading and preprocessing
NumPy	Tensor creation and numerical operations
Groq SDK	LLaMA 3.3-70b chat completion client
scikit-learn	SVC training (offline, models pre-exported)
TensorFlow/Keras	CNN model training (offline, models pre-exported)

python-dotenv	Environment variable management
httpx	Async HTTP client for external API calls

8.3 ONNX Models Summary

Model File	Task	Input Shape	Output	Accuracy
braintumor.onnx	Brain MRI classification	[1,168,168,1]	4-class softmax	~99%
lung.onnx	Chest X-Ray classification	[1,224,224,3]	Binary sigmoid	~90%+
skin.onnx	Skin condition detection	[1,64,64,1]	3-class softmax	~85%+
svc.onnx	Symptom -> Disease	[1,132]	41-class label	~94%

8.4 Infrastructure

Service	Purpose	Plan
Vercel	Frontend + Nitro SSR + Python serverless	Hobby/Pro
Appwrite	Authentication + Database collections	Cloud Free
Cloudflare	DNS management + local tunnel	Free
GitHub	Version control + CI/CD (auto-deploy)	Free
Docker	Containerized local + production deploy	Optional

9. DATABASE DESIGN

9.1 Appwrite Collections

Orders Collection Schema

```
Collection: orders
{
  "orderId":      "SS-20260907-A1B2C3D4" // auto-generated
  "userId":       "appwrite_user_id"
  "userName":     "Maryam Tahir"
  "userEmail":    "user@example.com"
  "items": [
    {
      "id":        "paracetamol_500mg",
      "name":      "Paracetamol 500mg",
      "qty":       2,
      "price":     150.0
    }
  ],
  "total":        300.0,
  "paymentMethod": "COD",
  "status":       "pending | confirmed | shipped | delivered",
  "estimatedDelivery": "2-3 business days",
  "createdAt":    "2026-09-07T20:00:00Z"
}
```

Reviews Collection Schema

```
Collection: reviews
{
  "userId": "appwrite_user_id",
  "userName": "Maryam T.",
  "rating": 5,
  "comment": "Excellent diagnostic accuracy!",
  "module": "diagnostics | symptoms | chat | prescription | pharmacy",
  "createdAt": "2026-09-07T20:00:00Z"
}
```

9.2 Local State (Browser)

Storage Key	Data	Type	Persistence
sehatsaathi_cart	Cart items array	CartItem[]	localStorage (permanent)
theme	Dark/light toggle	string	localStorage (permanent)
lang	EN/UR preference	string	localStorage (permanent)

9.3 CSV Dataset Files

File	Rows	Purpose
Medicine_Details.csv	15,000+	Full pharmacy database with brand, generic, price
description.csv	41	Disease descriptions for SVC results
precautions_df.csv	41	Precautions per disease (4 per disease)
medications.csv	41	Recommended medicines per disease
diets.csv	41	Diet plan per disease
workout_df.csv	41	Workout/activity recommendations per disease
advanced_cv_model.json	N/A	Centroid ML fallback model (JSON)

10. API SPECIFICATION

10.1 Base URLs

- Production: <https://sehatsaathi-ai.vercel.app>
- Local Development: <http://localhost:8000>
- Authentication: Appwrite cookie-based sessions (no API key required)

10.2 Endpoints

POST /api/assistant/chat

Purpose: AI Medical Chat (Groq LLaMA 3.3-70b)

Request:

```
{"message":"I have chest  
pain","language":"en","history":[{"role":"user","content":"Hello"}, {"role":"assista  
nt","content":"Hello! How can I help?"}]}
```

Response:

```
{"response":"Chest pain lasting more than a few minutes...","model":"llama-3.3-70b-  
versatile","language":"en","products":[{"id":"aspirin","name":"Aspirin  
100mg","price":120.0}]}
```

POST /analyze-medical-image

Purpose: Medical Image Analysis (ONNX)

Request:

```
multipart/form-data: file=<image_binary>, modality="auto"|"mri"|"xray"
```

Response:

```
{"modality":"mri","model_used":"High-Precision ONNX Engine","finding":"Glioma  
Tumor","confidence":0.924,"abnormality_score":0.88,"confidence_band":"High","full_c  
linical_report":{"disease":"Glioma  
Tumor","medicines":[...],"diet_plan":[...],"precautions":[...],"lab_tests":[...],"  
clinical_disclaimer":"..."}
```

POST /analyze-skin

Purpose: Skin Condition Analysis

Request:

```
multipart/form-data: file=<skin_image>
```

Response:

```
{"condition":"Acne","confidence":0.85,"skin_type":"Oily","recommendations":[...],"i  
ngredients":[...]}
```

POST /predict-disease

Purpose: Symptom-based Disease Prediction

Request:

```
{"symptoms":["headache","high_fever","nausea","vomiting"]}
```

Response:

```
{"disease":"Typhoid","description":"...", "precautions":["Drink boiled  
water","..."],"medicines":[{"name":"Ciprofloxacin 500mg"}],"diet_plan":["Light  
meals","..."],"workout_plan":["Bed rest"],"lab_tests":["Blood culture","Widal  
test"]}
```

GET /medicines/search?q={name}

Purpose: Medicine Search

Request:

```
Query param: q=paracetamol
```

Response:

```
{"name":"Paracetamol 500mg","generic":"Paracetamol","uses":"Pain and fever relief","price":120.0,"manufacturer":"GSK Pakistan","found":true}
```

GET /medicines/all?limit={n}**Purpose: List All Medicines (paginated)****Request:**

```
Query params: limit=40, offset=0
```

Response:

```
[{"id":"...", "name":"...", "price":..., "category":"...", "rating":4.5}]
```

POST /checkout**Purpose: Place COD Order****Request:**

```
{"items":[{"id":"med_1","name":"Paracetamol 500mg","qty":2,"price":150}], "total":300, "payment_method":"COD"}
```

Response:

```
{"status":"success","order_id":"SS-20260907-A1B2C3D4","message":"Order placed","payment_method":"Cash on Delivery","estimated_delivery":"2-3 business days"}
```

11. FRONTEND ARCHITECTURE

11.1 Global State (AppContext)

```
type AppState = {
  // Internationalization
  lang:          "en" | "ur"
  setLang:       (l: Lang) => void
  t:             (key: DictKey) => string // translation function
  rtl:          boolean                // auto-true for Urdu

  // Theme
  dark:         boolean
  toggleDark:   () => void

  // Shopping Cart
  cart:         CartItem[]
  cartCount:    number
  cartTotal:    number
  addToCart:    (item: CartItem) => void
  setQty:       (id: string, qty: number) => void
  removeFromCart: (id: string) => void
}
```

```

clearCart:      () => void
cartOpen:       boolean
setCartOpen:    (v: boolean) => void

// Authentication
user:           AppUser | null
authLoading:    boolean
authOpen:       boolean
setAuthOpen:    (v: boolean) => void
signIn:         (email: string, password: string) => Promise<void>
signUp:         (name: string, email: string, password: string) => Promise<void>
signOut:        () => Promise<void>
}

```

11.2 Route Structure

```

frontend/src/routes/
__root.tsx      (Root layout: Navbar, CartSheet, AuthDialog, Toaster)
index.tsx       -> / (Landing page)
diagnostics.tsx -> /diagnostics (AI medical image analysis)
symptoms.tsx    -> /symptoms (Symptom checker)
chat.tsx        -> /chat (AI assistant)
prescription.tsx -> /prescription (OCR scanner)
pharmacy.tsx    -> /pharmacy (Medicine store)
hospitals.tsx   -> /hospitals (Hospital locator)
diet.tsx        -> /diet (Diet planner)
admin.tsx       -> /admin (Admin dashboard)
about.tsx       -> /about
privacy.tsx     -> /privacy
terms.tsx       -> /terms

```

11.3 Internationalization (i18n)

- ~60 translation keys covering all UI text
- Supports: English (en) and Urdu (ur)
- RTL layout auto-applied via `document.documentElement.dir = "rtl"`
- Language persisted in `localStorage` and toggled via `NavBar`

12. SECURITY & AUTHENTICATION

12.1 Appwrite Auth Flow

```

Registration:
  authService.register(name, email, password)
  -> Appwrite creates account + email session

Login:
  authService.login(email, password)
  -> Appwrite sets HttpOnly cookie session

```

```
Session check:
  authService.getCurrentUser()
  -> Returns AppUser | null on every page load
```

```
Logout:
  authService.logout()
  -> Destroys server session, clears cookie
```

12.2 Admin Role Guard

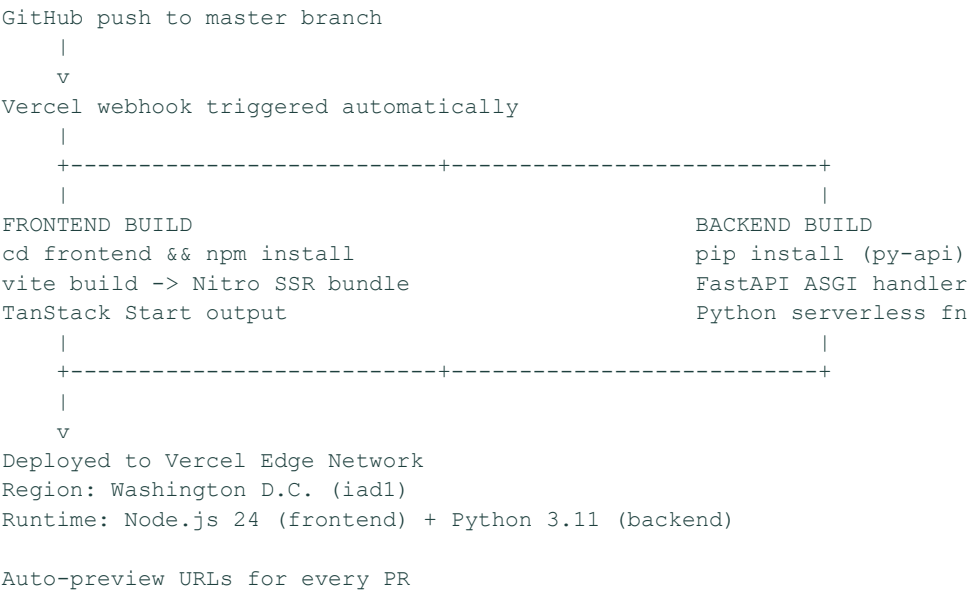
```
// admin.tsx
const { user, authLoading } = useAppContext()
if (!authLoading && (!user || !user.labels?.includes("admin"))) {
  return <Navigate to="/" />
}
```

12.3 Medical Safety Guardrails

Risk	Mitigation
AI makes wrong diagnosis	All outputs labeled as AI-generated, doctor required
AI prescribes medication	System prompt: "NEVER prescribe, always recommend doctor"
User data privacy (OCR)	Tesseract.js runs 100% client-side, no image sent to server
Model overconfidence	Confidence bands: Low (<70%), Medium (70-85%), High (>85%)
Definitive diagnosis claims	Disclaimer appended to every clinical report output

13. DEPLOYMENT ARCHITECTURE

13.1 Vercel CI/CD Pipeline



Production URL: sehat-saathi-ai.vercel.app

13.2 vercel.json Configuration

```
{
  "builds": [
    { "src": "frontend/package.json", "use": "@vercel/node" },
    { "src": "frontend/py-api/index.py", "use": "@vercel/python" }
  ],
  "routes": [
    { "src": "/api/(.*)", "dest": "frontend/py-api/index.py" },
    { "src": "/(.*)", "dest": "frontend/package.json" }
  ]
}
```

13.3 Environment Variables

Variable	Purpose	Used By
GROQ_API_KEY	Groq LLaMA chat	py-api
GROQ_MODEL	Model name config	py-api
GEMINI_API_KEY	Optional OCR fallback	py-api
NEXT_PUBLIC_APPWRITE_ENDPOINT	Appwrite server URL	frontend
NEXT_PUBLIC_APPWRITE_PROJECT_ID	Project identifier	frontend
NEXT_PUBLIC_APPWRITE_DATABASE_ID	DB identifier	frontend
NEXT_PUBLIC_APPWRITE_BUCKET_ID	Storage bucket	frontend
NEXT_PUBLIC_API_URL	Backend API base URL	frontend

14. PERFORMANCE & CONSTRAINTS

14.1 Vercel Serverless Limits

Limit	Vercel Value	Our Approach
Bundle size	250 MB	ONNX models (not PyTorch/TF) — 10-50x smaller
Execution timeout	60 seconds	ONNX inference < 3s, chat < 5s
Memory limit	1024 MB	Lightweight PIL + ONNX, no heavy frameworks
Cold start	1-3 sec	Model loaded once, cached for session
File size per fn	250 MB	Total py-api bundle < 80MB

14.2 Model Inference Performance

Model	Inference Time	Memory	Training Accuracy
braintumor.onnx	~300ms	~45MB	~99%
lung.onnx	~200ms	~30MB	~90%+
skin.onnx	~150ms	~20MB	~85%+
svc.onnx	<10ms	<1MB	~94%

Centroid ML JSON	<1ms	<100KB	N/A (heuristic)
------------------	------	--------	-----------------

14.3 Frontend Performance

- Code-split routes via Vite — only current page JS loaded
- Framer Motion animations GPU-accelerated (transform, opacity only)
- Cart persisted in localStorage — zero API calls on load
- Tesseract.js worker spawned once, reused across scans
- React 19 concurrent features for non-blocking UI updates

15. FUTURE ROADMAP

Version	Feature	Description	Timeline
v1.1	Voice Input	Web Speech API for hands-free medical queries	Q4 2026
v1.1	WhatsApp Bot	WABA integration for rural feature-phone users	Q4 2026
v1.1	Appointment Booking	Doctor slot booking with calendar integration	Q4 2026
v1.2	Retinal Scan AI	Diabetic retinopathy detection from fundus images	Q1 2027
v1.2	Blood Report OCR	Parse CBC/LFT/KFT from lab report photos	Q1 2027
v1.2	Telemedicine	WebRTC video calls with verified doctors	Q1 2027
v2.0	Mobile App	React Native iOS + Android application	Q3 2027
v2.0	Wearables	Apple Watch / Fitbit vitals integration	Q3 2027
v2.0	Govt. Hospital API	Pakistan DHQ / PMDC hospital data integration	Q3 2027
v2.0	Insurance Claims	AI-assisted claim form auto-fill + submission	Q3 2027

SehatSaathi AI

Making healthcare intelligent, accessible, and human
for every Pakistani.

github.com/maryamtahir7/SehatSaathi-AI
sehatsaathi-ai.vercel.app

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